

OBAN, United Kingdom

WMO: 031140

Lat: 56.417N

Lon: 5.467W

Elev: 13

StdP: 14.69

Time Zone: 0.00 (EUW)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	28.0	30.7	21.2	15.9	34.9	23.6	17.8	34.9	34.6	41.1	30.1	45.1	0.5	160	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	74.0	62.0	70.4	60.2	67.3	58.9	63.9	71.1	61.9	68.1	60.2	65.3	6.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	79.8	67.5	59.2	75.1	64.3	57.8	71.4	62.8	29.1	71.6	27.7	68.2	26.5	65.3	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.7	22.8	19.5	DB	23.7	78.7	3.8	3.0	21.0	80.9	18.8	82.6	16.7	84.3	13.9	86.5	(4)
(5)			WB	22.9	68.1	3.6	5.5	20.3	72.1	18.2	75.3	16.2	78.3	13.6	82.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	41.7	41.8	43.8	46.4	52.8	55.3	59.1	59.1	54.8	49.9	45.5	41.1
	DBStd	7.93	4.40	4.59	4.04	4.14	5.22	3.87	4.21	4.42	4.08	4.72	5.09	5.63
	HDD50	1323	258	230	193	120	27	2	0	0	6	61	148	278
	HDD65	5743	722	649	657	557	378	293	191	196	306	468	585	741
	CDD50	1080	1	1	1	13	115	161	283	281	151	58	13	2
	CDD65	23	0	0	0	0	1	2	8	12	0	0	0	0
	CDH74	96	0	0	0	0	3	13	45	35	0	0	0	0
	CDH80	4	0	0	0	0	0	1	1	2	0	0	0	0
Wind	WSAvg	6.2	8.3	8.3	7.3	6.2	5.2	5.1	4.6	4.4	5.0	6.4	7.4	6.9
Precipitation	PrecAvg	84.7	10.7	7.7	7.2	5.0	4.6	4.6	5.0	5.7	6.8	8.9	9.0	10.4
	PrecMax	112.0	18.6	16.0	17.8	8.6	11.8	8.7	8.0	11.7	13.1	15.0	14.1	18.6
	PrecMin	57.0	4.0	2.8	1.2	1.7	1.3	1.6	2.2	1.6	1.7	2.0	3.9	2.2
	PrecStd	13.7	3.5	4.1	3.7	1.6	2.5	1.8	1.4	2.4	3.0	3.9	3.0	4.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	53.8	54.3	62.3	74.1	75.3	78.4	77.8	69.7	63.1	57.8	54.5
		MCWB	50.0	51.7	48.4	52.9	58.5	59.8	65.2	64.5	62.4	56.4	54.7	52.1
	2%	DB	51.6	51.9	51.9	57.4	69.6	69.1	73.9	73.6	65.5	60.0	55.6	52.5
		MCWB	49.6	49.5	48.1	49.6	55.7	58.4	63.1	63.1	58.7	55.3	52.4	50.4
	5%	DB	49.7	50.2	50.5	54.7	65.4	65.2	69.5	69.7	63.1	58.3	54.3	50.6
		MCWB	47.5	48.2	47.5	48.7	54.5	56.6	60.2	62.0	57.2	54.5	51.6	48.4
	10%	DB	48.0	48.5	49.4	52.7	61.6	62.1	66.2	66.2	61.3	56.7	52.7	48.9
		MCWB	45.9	46.6	46.8	47.6	52.7	54.7	58.6	59.6	56.3	53.2	50.1	46.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	52.5	50.5	53.9	59.5	62.9	67.3	66.2	63.5	57.6	55.5	52.8
		MCDB	53.0	53.3	52.1	59.1	71.0	72.7	74.7	74.9	68.1	60.6	57.1	54.1
	2%	WB	49.7	49.8	49.3	51.6	57.1	59.7	63.7	64.3	60.5	56.5	53.6	50.8
		MCDB	51.3	51.1	50.6	55.9	67.5	66.4	71.3	71.6	64.0	59.1	55.1	52.3
	5%	WB	47.7	48.4	48.4	49.9	55.3	57.7	61.4	62.4	58.5	55.2	51.9	48.7
		MCDB	49.2	49.9	49.7	53.3	63.2	63.3	67.6	68.6	61.7	57.8	53.9	50.2
	10%	WB	46.3	46.9	47.5	48.6	53.6	56.0	59.6	60.7	57.0	53.7	50.2	47.0
		MCDB	47.9	48.5	49.1	51.5	60.2	60.9	65.0	65.4	60.4	56.4	52.6	48.9
Mean Daily Temperature Range	5% DB	MDBR	7.0	7.0	7.6	9.4	12.7	10.6	10.1	10.4	10.7	8.3	7.4	7.4
		MCDBR	8.3	6.6	8.5	12.8	19.5	16.5	17.4	15.2	12.8	9.0	7.4	8.0
		MCWBR	8.2	6.5	6.4	8.1	10.3	9.0	9.4	8.0	8.0	7.2	7.1	8.4
	5% WB	MCDBR	7.8	6.2	6.8	10.9	17.3	14.5	14.5	13.6	10.8	7.7	7.0	8.0
		MCWBR	8.0	6.5	6.4	8.1	9.9	8.7	8.8	8.5	7.7	7.6	7.7	8.5
Clear-Sky Solar Irradiance	taub	0.338	0.348	0.368	0.382	0.387	0.379	0.383	0.366	0.362	0.346	0.328	0.323	
	taud	2.179	2.280	2.290	2.301	2.336	2.394	2.392	2.449	2.456	2.412	2.270	2.152	
	Ebn at Noon	165	221	249	264	270	274	270	268	251	222	171	137	
	Edh at Noon	19	25	31	36	36	35	34	30	26	22	17	15	
All-Sky Solar Radiation	RadAvg	133	334	664	1104	1442	1401	1340	1064	731	391	174	101	
	RadStd	21	47	57	89	135	158	137	96	60	54	16	14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page